HW 12: Class12 Pt.2 (Population analysis)

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```r
x <- read.table("BIMM143HW12.txt", header = TRUE)

head(x)
```

```
   sample geno      exp
1 HG00367  A/G 28.96038
2 NA20768  A/G 20.24449
3 HG00361  A/A 31.32628
4 HG00135  A/A 34.11169
5 NA18870  G/G 18.25141
6 NA11993  A/A 32.89721
```

```r
colnames(x) <- c("sample", "genotype", "expression")
colnames(x)
```

```
[1] "sample"     "genotype"   "expression"
```

> Q13: Read this file into R and determine the sample size for each genotype and their corresponding median expression levels for each of these genotypes.

Sample size for each genotype

```r
table(x$genotype)
```

```

A/A A/G G/G 
108 233 121 
```

Median expression level for each genotype

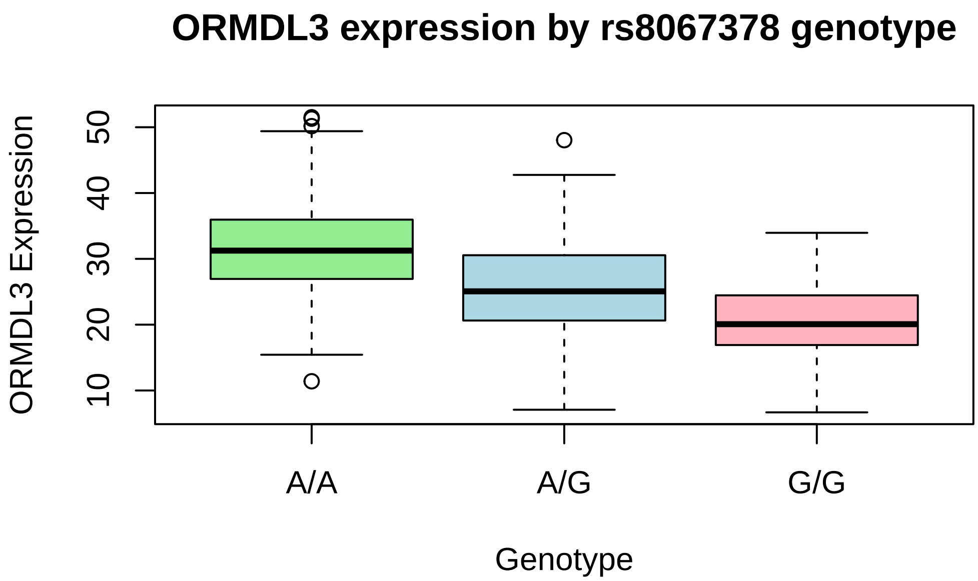
```r
tapply(x$expression, x$genotype, median)
```

```
     A/A      A/G      G/G 
31.24847 25.06486 20.07363 
```

> Q14: Generate a boxplot with a box per genotype, what could you infer from the relative expression value between A/A and G/G displayed in this plot? Does the SNP effect the expression of ORMDL3?

The boxplot shows that individuals with the A/A genotype have higher ORMDL3 expression levels than individuals with the G/G genotype. The G/G group has the lowest median expression, while the A/A group has the highest. This suggests that the rs8067378 SNP is associated with differences in ORMDL3 expression and likely affects the expression of the ORMDL3 gene (see below).

```r
boxplot(expression ~ genotype, data = x,
        xlab = "Genotype",
        ylab = "ORMDL3 Expression",
        main = "ORMDL3 expression by rs8067378 genotype",
        col = c("lightgreen", "lightblue", "lightpink"))
```


ORMDL3 expression by rs8067378 genotype
ORMDL3 Expression
10
20
30
40
50
A/A
A/G
G/G
Genotype